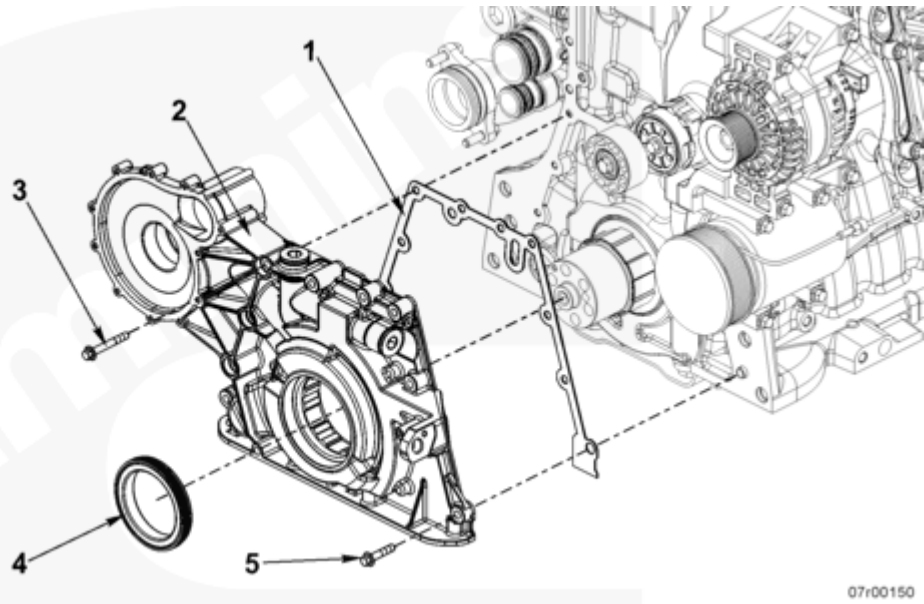


Trainings ▾

Exploded View

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Lubricating Oil Pump Exploded View

1. Gasket - quantity 1
2. Lubricating oil pump - quantity 1
3. Capscrew, long - quantity 5
4. Oil seal - quantity 1
5. Capscrew, short - quantity 6

Select Service Tools

Recommended Cummins® Service Tools

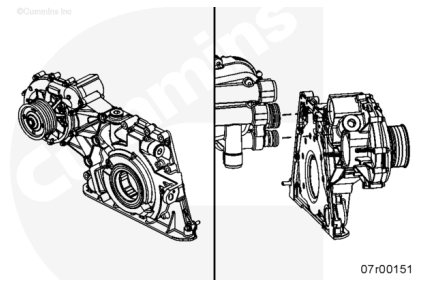
- Guide pins, Part Number 3164977

Additional Service Items

- No additional service items required.

General Information

This engine uses a rotor-style lubricating oil pump that is powered directly from the crankshaft nose. The water pump housing, lubricating oil pump, lubricating oil high-pressure relief valve, and lubricating oil rifle pressure regulator are all part of the lubricating pump housing assembly.



The lubricating oil pump rotor is **not** serviceable and the pump assembly shall **not** be disassembled. If any part of the lubricating oil pump assembly malfunctions, the entire lubricating pump assembly **must** be replaced. The lubricating oil high-pressure relief valve and lubricating oil rifle pressure regulator are serviceable and may be removed separately.

Use the following procedure for more information on the lubricating oil high-pressure relief valve. Refer to Procedure 007-021 in Section 7.

(/qs3/pubsys2/xml/en/procedures/1016/1016-007-021.html)

Use the following procedure for more information on the lubricating oil pressure regulator. Refer to Procedure 007-029 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-029.html)

The lubricating oil pump is connected to the lubricating oil cooler module with an oil transfer tube and a coolant transfer tube. The oil transfer tube and coolant transfer tube allow for the module and pump to be removed independently of each other.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

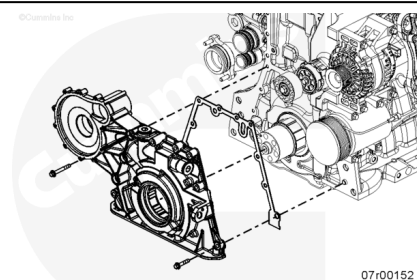
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/1016/1016-013-005-tr.html)
- Remove the cooling fan drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-002-tr.html)

- Remove the cooling fan belt tensioner. Refer to Procedure 008-087 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-087-tr.html>)
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-052.html>)
- Remove the crankshaft pulley and speed indicator ring. Refer to Procedure 001-022 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-022-tr.html>)
- Remove the fan pulley. Refer to Procedure 008-089 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-089-tr.html>)
- Disconnect and remove the crankshaft position/speed sensor. Refer to Procedure 019-365 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-365.html>)
- Remove the water pump cartridge, if required. Refer to Procedure 008-102 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-102.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)
- Remove the front engine support. Refer to Procedure 016-002 in Section 16. (</qs3/pubsys2/xml/en/procedures/1016/1016-016-002.html>)

Remove

⚠ CAUTION ⚠

To break the seal, pry the lubricating pump assembly away from the cylinder block. Be careful not to damage the lubricating pump assembly when breaking the seal to the cylinder block.



Note : Document the location of the mounting capscrews and the lubricating oil pump housing capscrews upon removal.

Remove the lubricating oil pump housing mounting capscrews.

Remove the lubricating oil pump assembly.

Disassemble

Remove the crankshaft oil seal.

Note : It is not necessary to remove the lubricating oil pressure regulator and lubricating oil high-pressure

relief valve unless there are signs or symptoms of damage. The lubricating oil pump may be cleaned with solvent or steam without removing the lubricating oil pressure regulator and lubricating oil high-pressure relief valve.

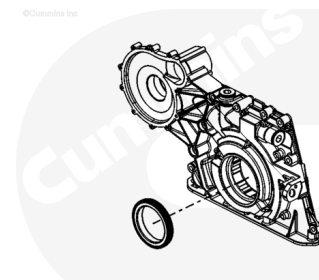
If necessary, remove the lubricating oil pressure regulator.

Refer to Procedure 007-029 in Section 7.

(/qs3/pubsys2/xml/en/procedures/1016/1016-007-029.html)

If necessary, remove the lubricating oil high-pressure relief valve. Refer to Procedure 007-021 in Section 7.

(/qs3/pubsys2/xml/en/procedures/1016/1016-007-021.html)

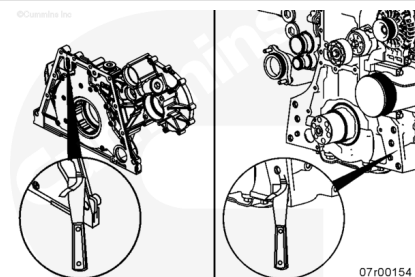


07r00153

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



07r00154

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

If the lubricating pump assembly requires steam cleaning, remove the lubricating oil high-pressure relief valve and oil rifle pressure regulator. Refer to Procedure 007-021 in Section 7 (/qs3/pubsys2/xml/en/procedures/1016/1016-007-021.html). Refer to Procedure 007-029 in Section 7. (/qs3/pubsys2/xml/en/procedures/1016/1016-007-029.html)

Note : If the lubricating pump assembly is removed as part of another repair, it is **only** necessary to clean the sealing surfaces.

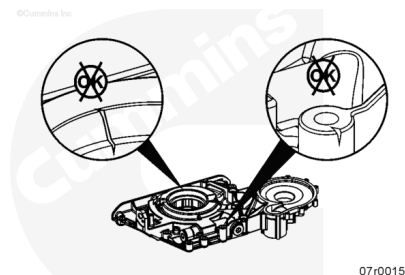
Clean the sealant or gasket from the lubricating pump assembly and from the cylinder block.

Use solvent or steam to clean the lubricating pump assembly.

Dry the assembly with compressed air.

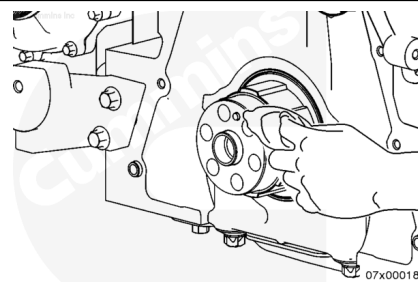
Inspect the lubricating pump housing for cracks or other damage.

Replace the lubricating pump assembly if any damage is found.



Use solvent to clean the oil and seal residue from the crankshaft surface.

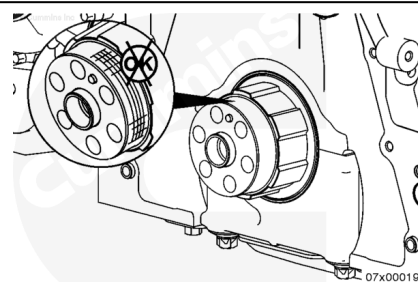
Dry with compressed air.



Inspect the nose of the crankshaft for excessive wear.

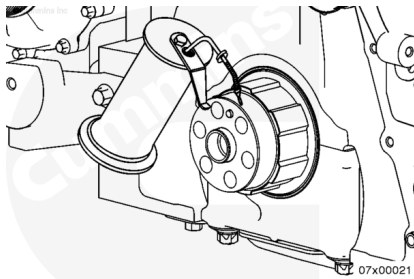
Use a fine crocus cloth to remove any nicks or burrs.

Use the following procedure for further inspection of the crankshaft nose. Refer to Procedure 001-016 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-016.html)



Install

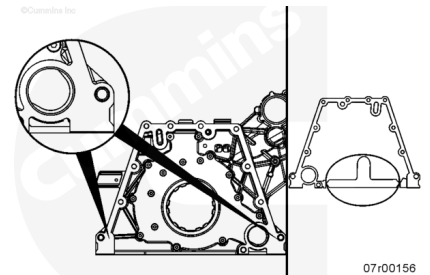
Lubricate the outside surface of the camshaft nose with clean engine oil.



When installing the lubricating pump housing gasket, do **not** damage the rubber seal on the bottom edges of the gasket.

Note : This gasket features pre-installed sealant to seal the joint between the lubricating oil pump housing, engine block, and lubricating oil pan. If this sealant is damaged, the gasket must be replaced.

Note : When installing a new gasket, remove the bridge first.



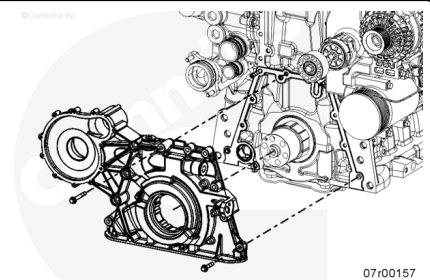
Install guide pins, Part Number 3164977, in the top corner capscrew locations as indicated.

Install the lubricating oil pump gasket. Use the guide pins and dowel rings in the cylinder block to locate and temporarily hold the lubricating oil pump gasket in place.

Note : If needed, rotate the lubricating oil pump in the lubricating pump housing to align with the lubricating oil pump drive gear on the crankshaft.

Install the lubricating oil pump.

Note : As documented during removal, install the mounting capscrews in the correct locations.

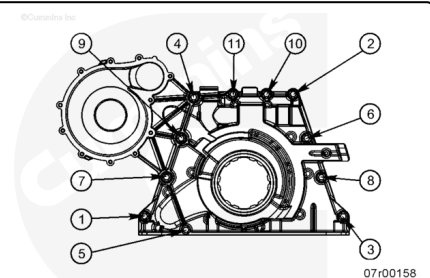


Install the lubricating oil transfer tube and coolant transfer tube into the lubricating oil pump. Verify correct length fasteners are used in the correct location. Tighten the capscrews finger-tight.

Remove the guide pins and install the two remaining capscrews.

Tighten the capscrews in the sequence shown.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the front engine support. Refer to Procedure 016-002 in Section 16.
(/qs3/pubsys2/xml/en/procedures/1016/1016-016-002.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Install and connect the crankshaft position/speed sensor. Refer to Procedure 019-365 in Section 19. (/qs3/pubsys2/xml/en/procedures/1016/1016-019-365.html)
- Install the front crankshaft lubricating oil seal. Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-023.html)
- Install the fan pulley. Refer to Procedure 008-089 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-089-tr.html)
- Install the crankshaft pulley and speed indicator ring. Refer to Procedure 001-022 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-022-tr.html)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-052.html)
- Install the cooling fan belt tensioner. Refer to Procedure 008-087 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-087-tr.html)
- Install the cooling fan drive belt. Refer to Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-002-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/1016/1016-013-005-tr.html)
- Install the water pump cartridge, if removed. Refer to Procedure 008-102 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-102.html)
- Prime the lubricating oil system. Refer to Procedure 014-005 in Section 14.
(/qs3/pubsys2/xml/en/procedures/1016/1016-014-005.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks and proper operation.